



CPG to Actionable Care Plans

Bringing actionable intelligence to
healthcare on Red Hat AI

Sam Schiffman



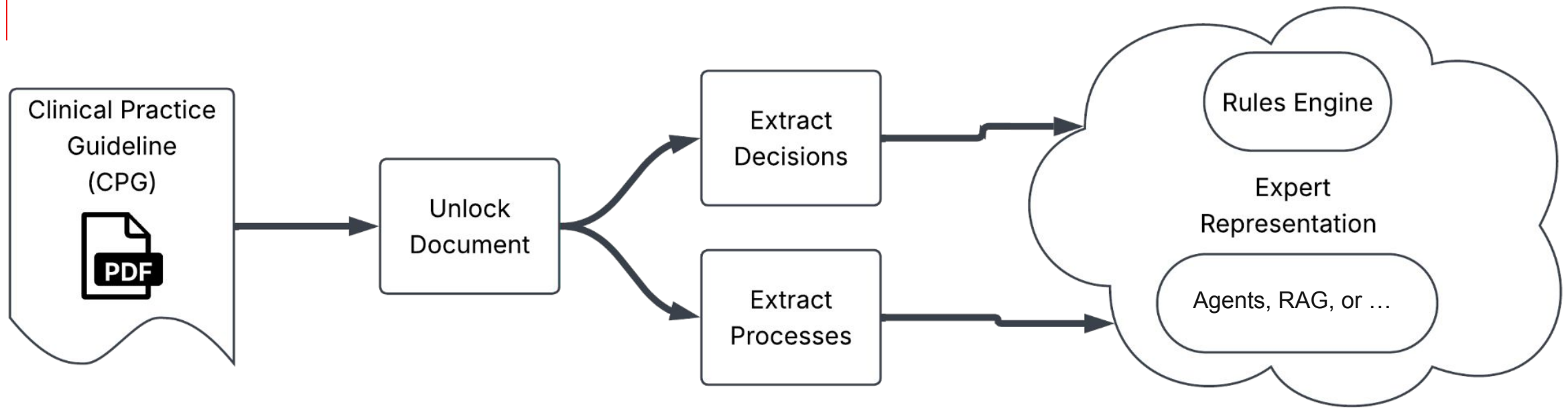
Imagine a patient comes into the Emergency Room. The doctor only has minutes to put together their history and diagnose the current problem. Then they have to build an effective plan for the patient's care.

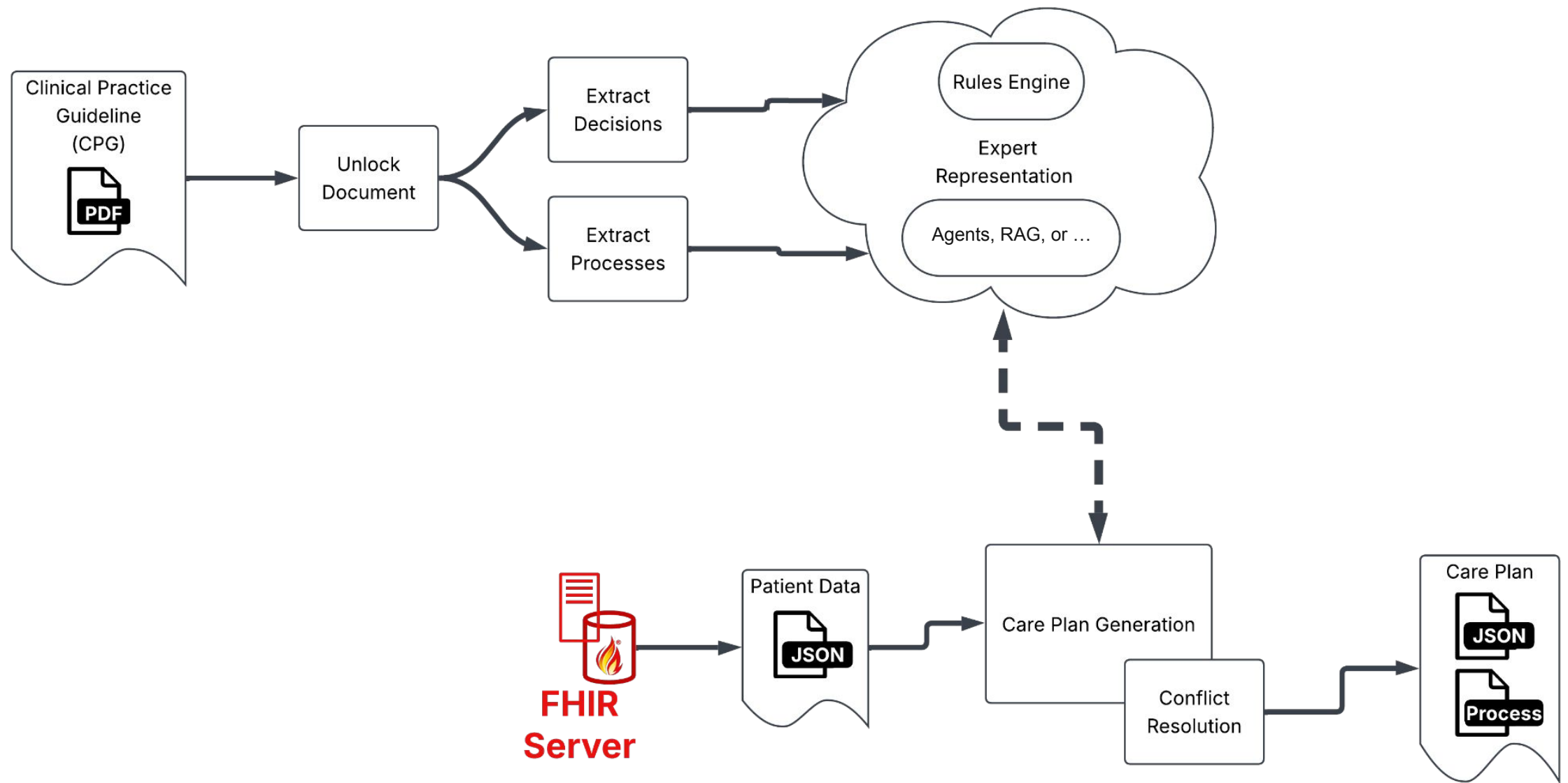
What if we could bring teams of experts right to the doctor's side? Suggest to a doctor what the care plan might be, based on the highest standard of care?

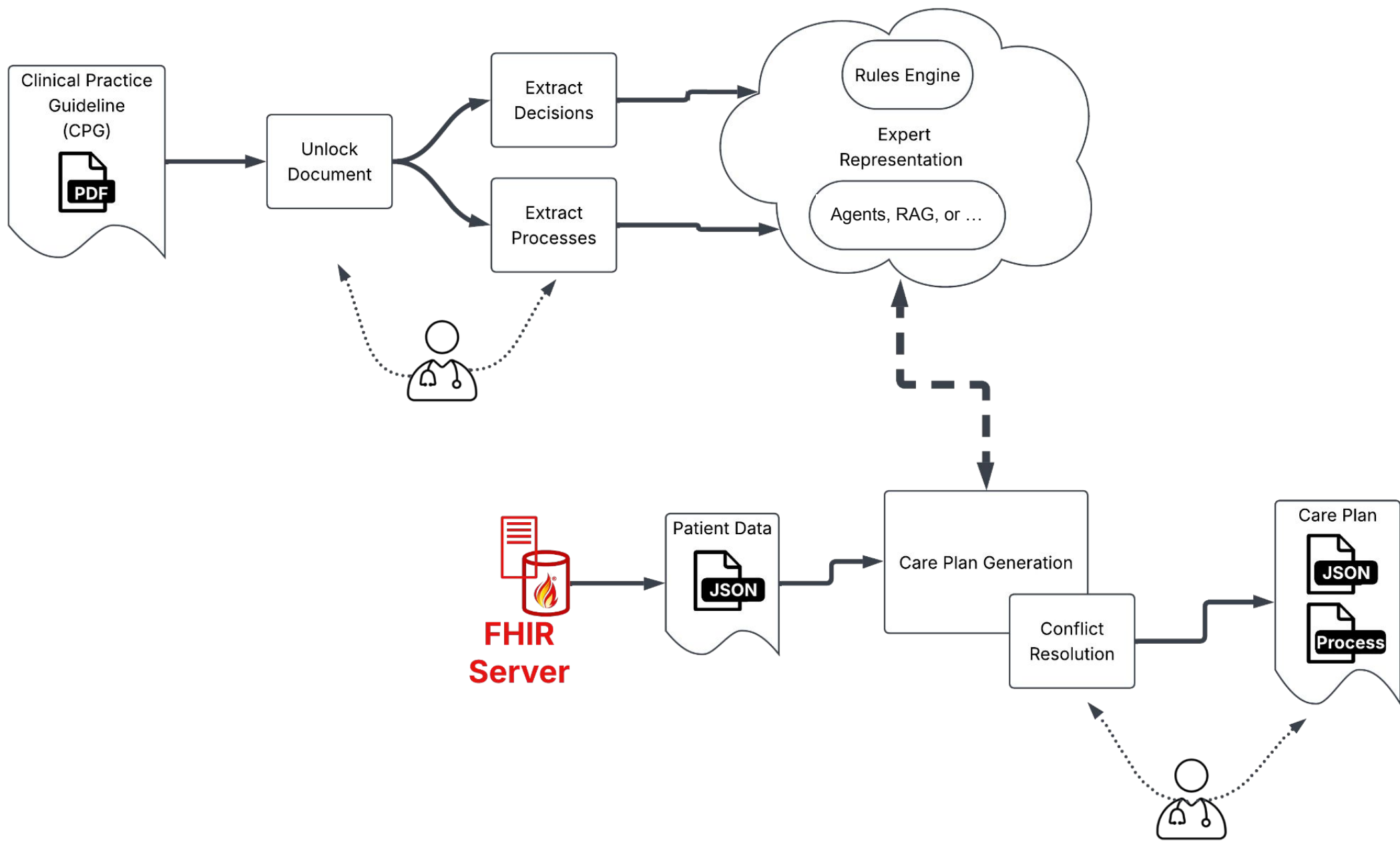
But beyond that, what if the suggestions include automations that could handle appropriate portions of the care?

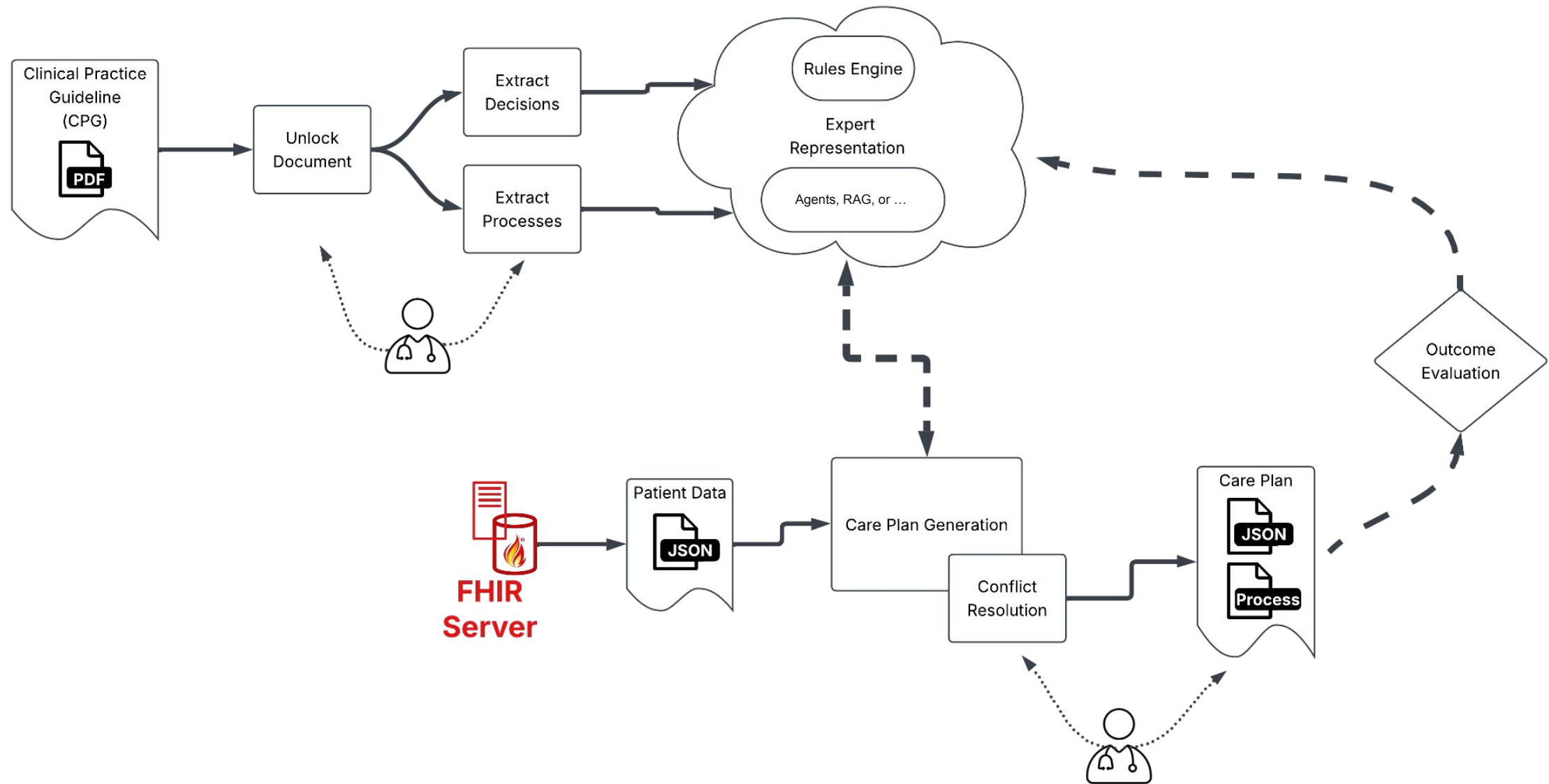
This would be an Actionable Care Plan the doctor could modify, approve, and put into motion right away.

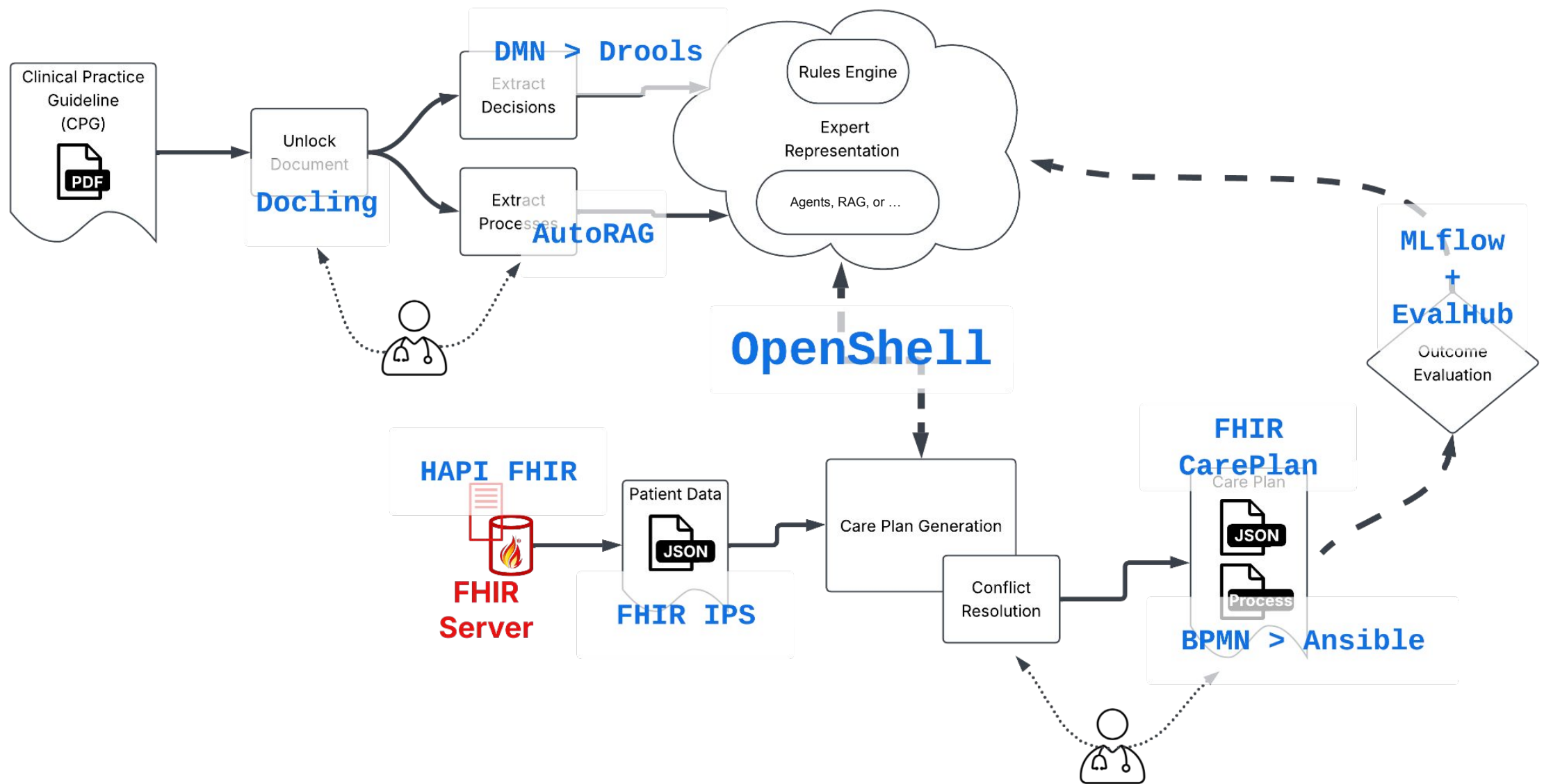






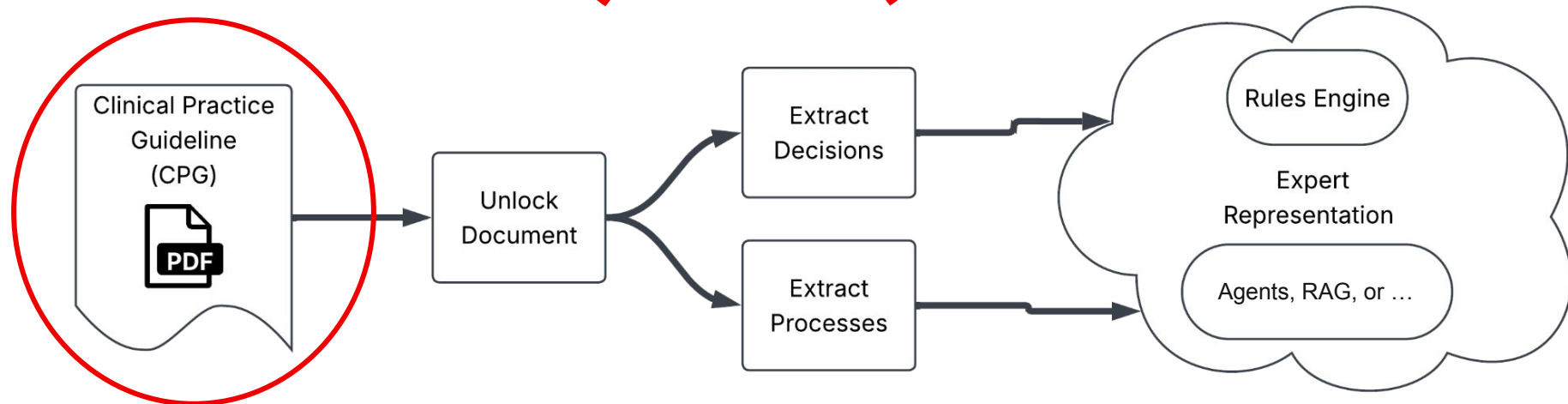




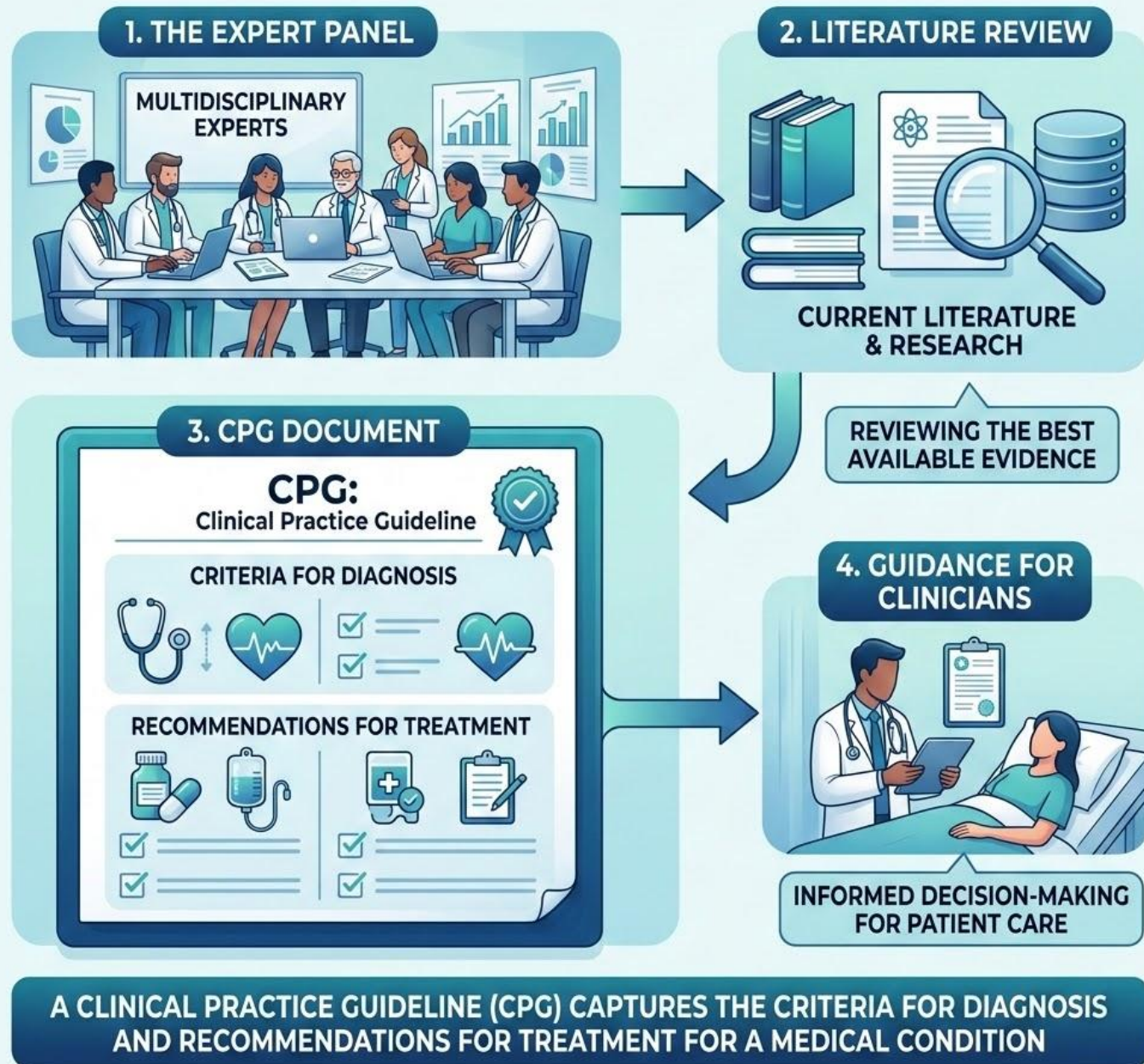


CPG to Actionable Care Plans

What is a Clinical Practice Guideline (CPG)

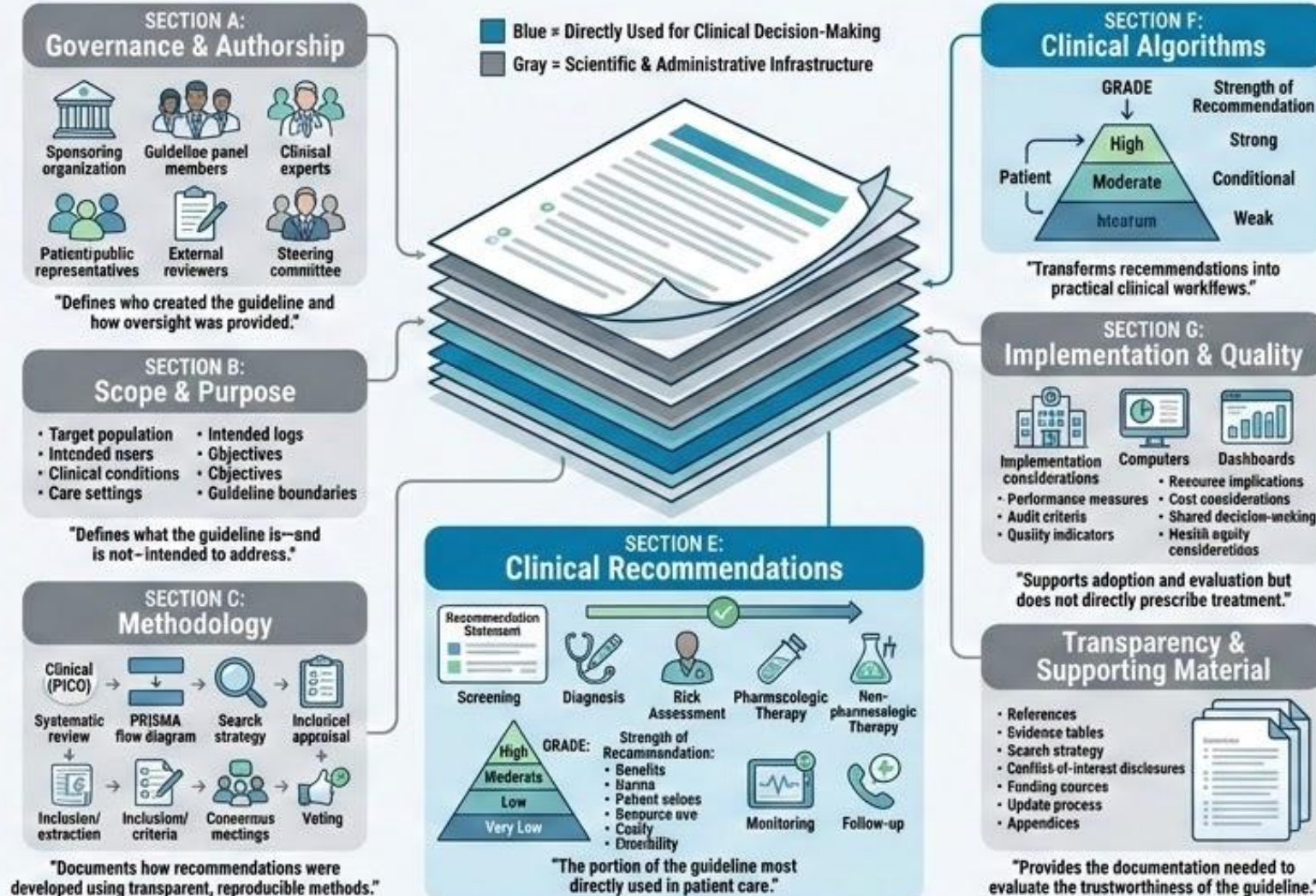


WHAT IS A CLINICAL PRACTICE GUIDELINE (CPG)?



Inside a Clinical Practice Guideline (CPG)

A Clinical Practice Guideline is more than a list of treatment recommendations. It combines **evidence synthesis**, **expert governance**, **transparent methodology**, and **clinical guidance** into a single structured document.



Scientific Infrastructure

(Authorship, Governance, Methodology, Evidence Review, Transparency, Implementation)

Clinical Guidance

(Recommendations, Algorithms, Decision Support)

Freely Available Clinical Practice Guidelines (CPGs)

Access authoritative, open-access medical recommendations and structured logic engines from global health systems.

US Dept. of Veterans Affairs

The VA/DoD publishes evidence-based Clinical Practice Guidelines to reduce variation in military and veteran health outcomes.

- Chronic Disease Management
- Mental Health & PTSD Care
- Rehabilitation & Pain Therapy

healthquality.va.gov ↗

World Health Organization

WHO guidelines provide global recommendations on public health interventions, disease prevention, and health system standards.

- Global Health Standards
- Infectious Disease Protocols
- Maternal & Child Health

who.int/publications/who-guidelines ↗

American College of Physicians

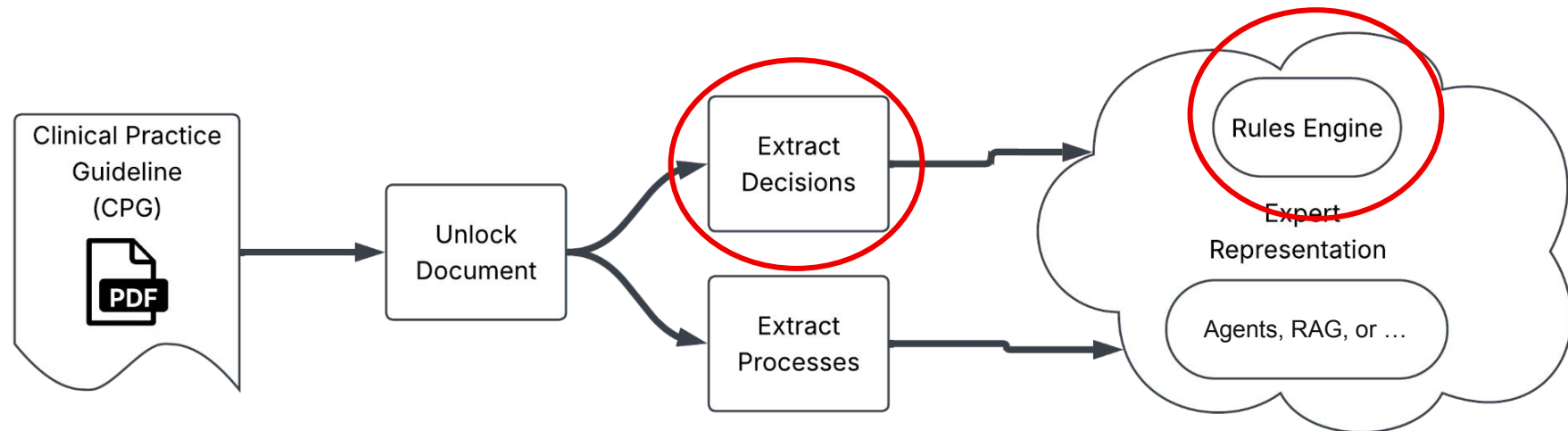
The American College of Physicians is the largest medical specialty organization in the United States with members in more than 172 countries worldwide. Our 163,000 members include internal medicine physicians, related subspecialists, and medical students.

acponline.org/clinical-information/clinical-guidelines-recommendations ↗



CPG to Actionable Care Plans

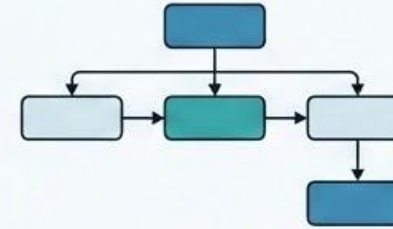
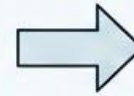
Extracting Decisions



What is Decision Model and Notation (DMN)?

A visual language for describing decisions and the rules behind them.

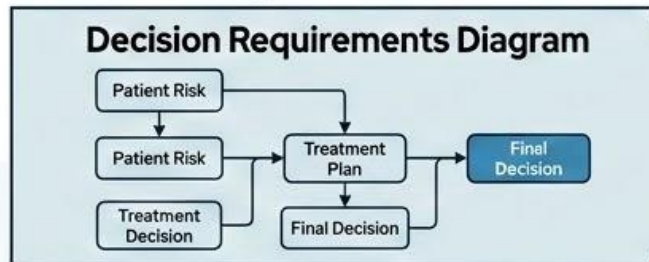
Why DMN?



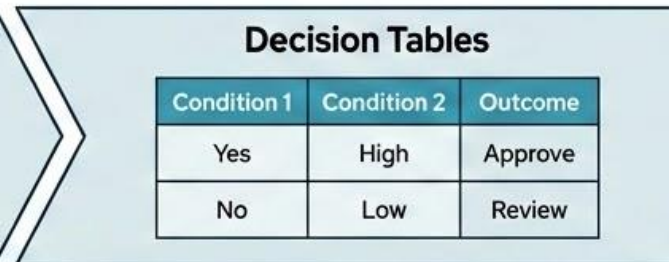
Decision Table		
Yes	Yes	Approve
No	High	Review
	Low	

DMN transforms complex written rules into clear visual models that are easier to review and understand.

The Building Blocks



Shows how decisions depend on one another.



Organizes rules into an easy-to-read format.



Captures the policies, guidelines, and expertise behind each decision.



Knowledge & Policies



Visual Decision Model



Shared Understanding



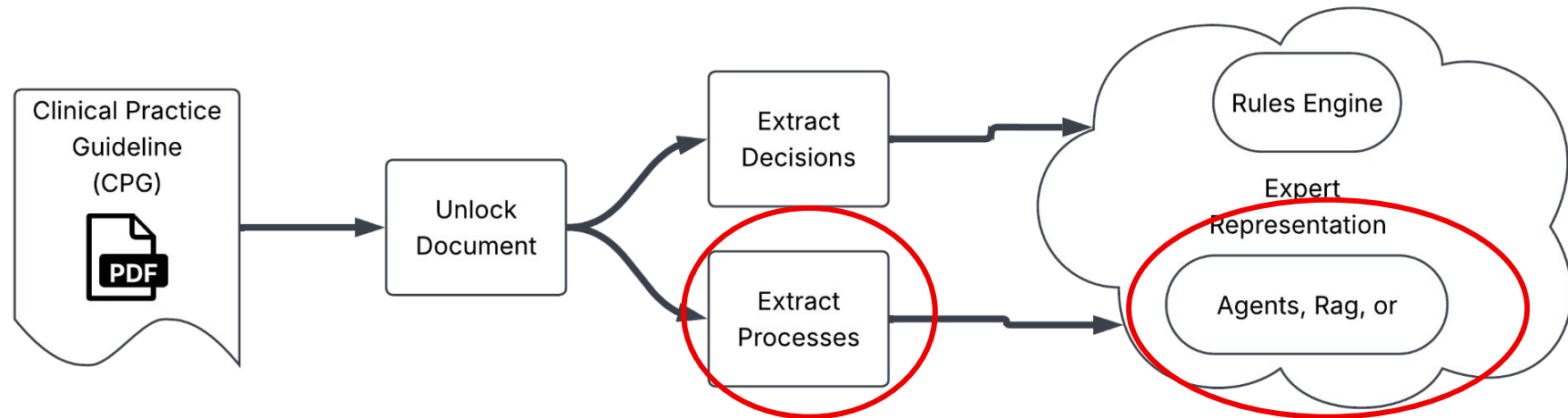
Better Decisions

DMN provides a common visual language that helps subject matter experts, analysts, and developers review decision logic together—making complex decisions easier to understand, validate, and maintain.

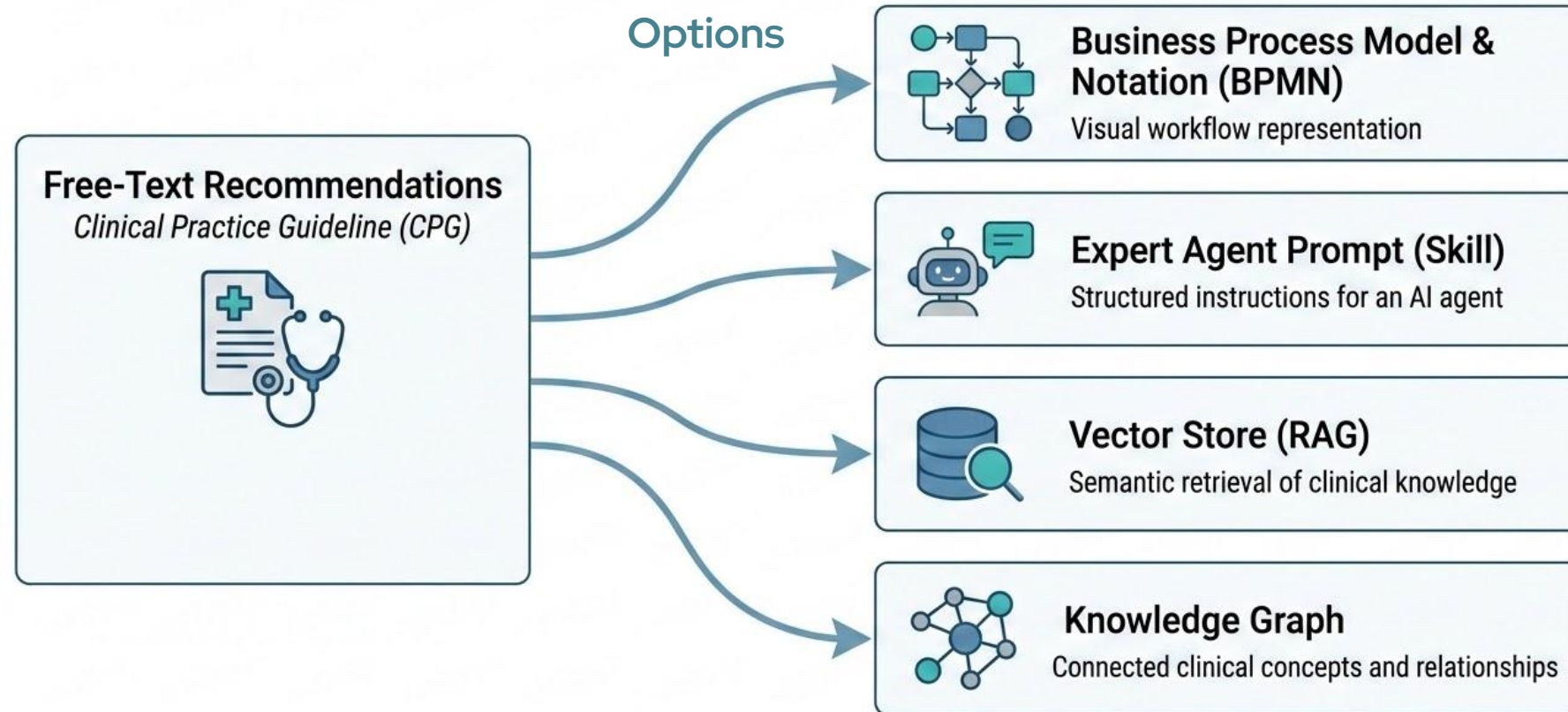


CPG to Actionable Care Plans

Extracting Process

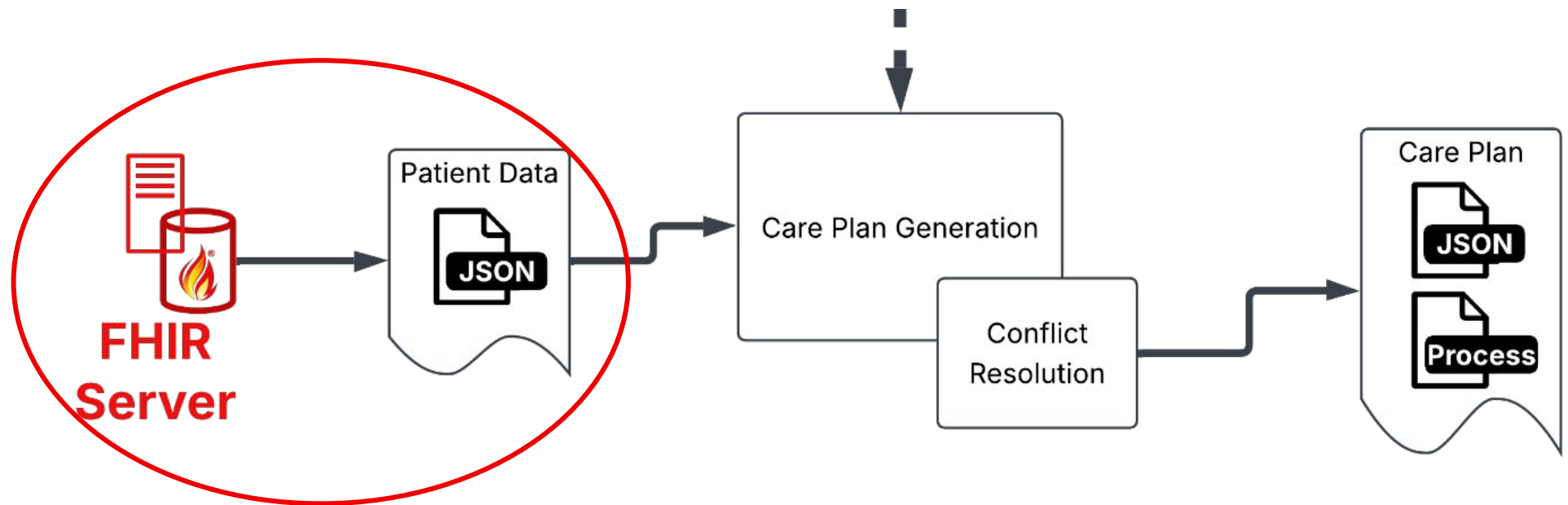


Transforming Clinical Practice Guideline (CPG) Free-Text Recommendations



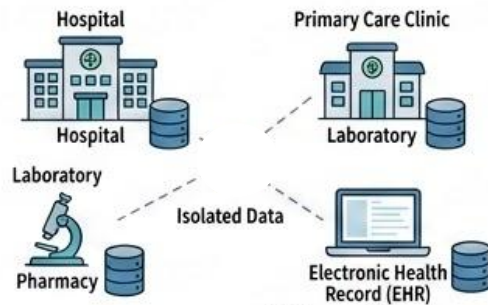
CPG to Actionable Care Plans

Patient Data



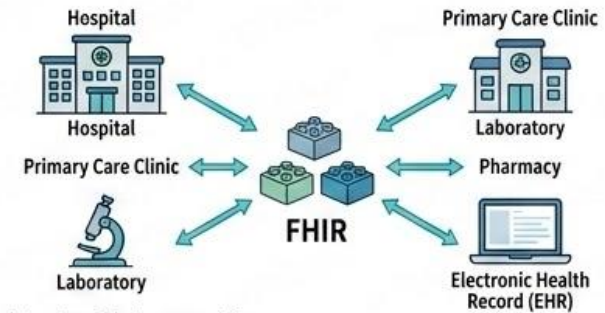
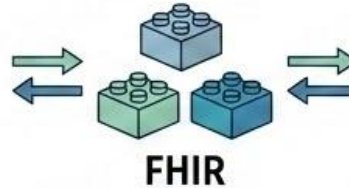
Fast Healthcare Interoperability Resources (FHIR)

A modern standard for sharing healthcare information across systems.

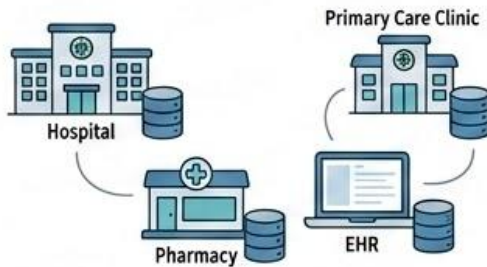


FHIR provides a common standard that allows healthcare systems to exchange information in a consistent and interoperable way.

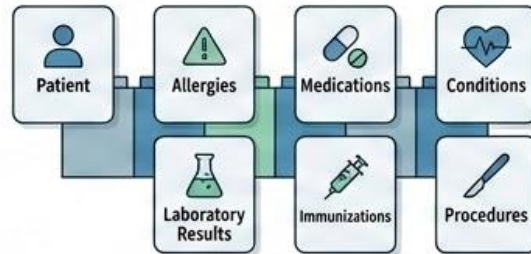
1 Why FHIR?



2 FHIR Organizes Healthcare Information into Resources

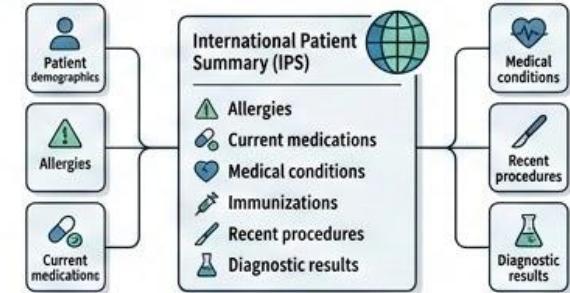


FHIR provides a common standard that allows healthcare systems to exchange information in a consistent and interoperable way.



FHIR organizes healthcare information into reusable "resources" that can be combined to describe a patient's health record.

3 International Patient Summary (IPS)



The International Patient Summary (IPS) is a standardized collection of essential health information designed to support safe patient care across organizations and countries.

Healthcare Systems

FHIR Resources

International Patient Summary (IPS)

Shared Clinical Understanding

FHIR provides the common language for healthcare data, while the International Patient Summary (IPS) defines a standardized clinical summary that can be shared wherever a patient receives care.



<https://hl7.org/fhir/>



HomeGetting StartedDocumentationData Types**Resource Types**TerminologiesArtifactsImplementation Guides

Home

This page is part of the FHIR Specification (v5.0.0: R5 - [STU](#)). This is the current published version. For a full list of available versions, see the [Directory of published versions](#).
Page versions: [R5](#) [R4B](#) [R4](#) [R3](#) [R2](#)

0 Welcome to FHIR®

FHIR is a standard for health care data exchange, published by HL7®.

First time here?
See the [executive summary](#), the [developer's introduction](#), [clinical introduction](#), [patient introduction](#), or [architect's introduction](#), and then the FHIR [overview & how FHIR versions work](#). See also the [open license \(CC0\)](#) (and don't miss the full [Table of Contents](#) and the [Community Credits](#) or you can [search this specification](#)).

Level 1 Basic framework on which the specification is built

 Foundation	Base Documentation, XML, JSON, RDF, Datatypes, Extensions
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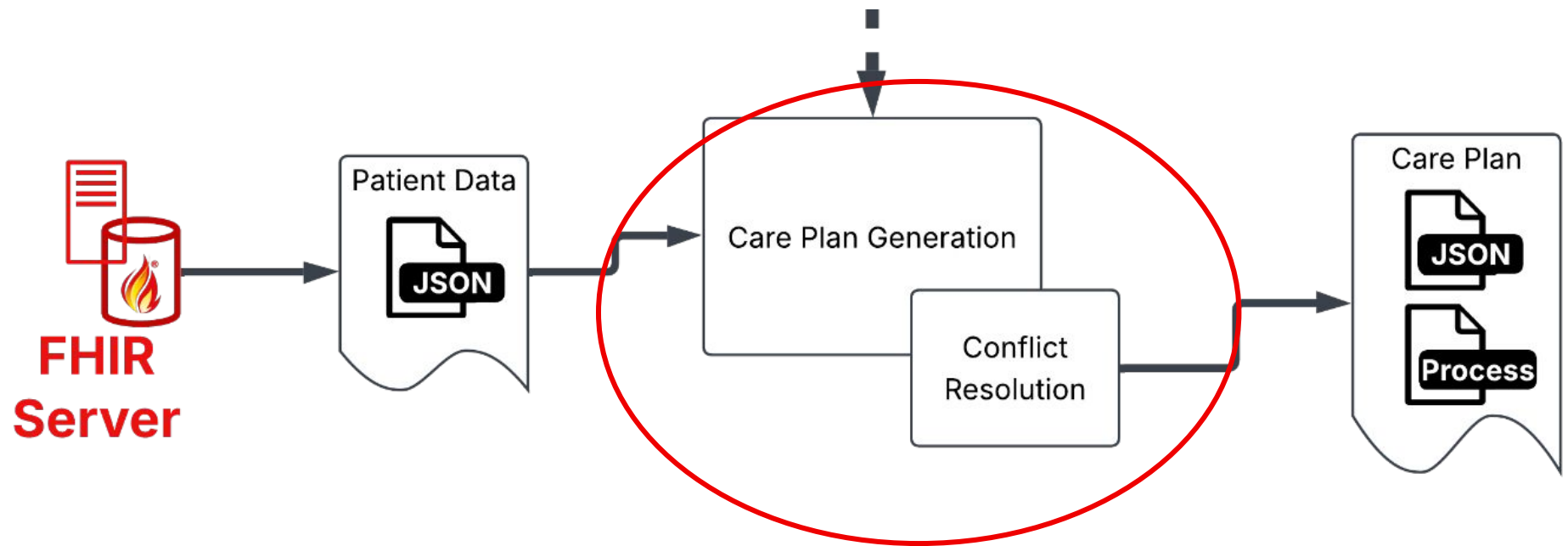
Level 2 Supporting implementation and binding to external specifications

 Implementer Support Downloads, Version Mgmt,	 Security & Privacy Security, Consent,	 Conformance StructureDefinition, CapabilityStatement,	 Terminology CodeSystem, ValueSet,	 Exchange REST API + Search Documents
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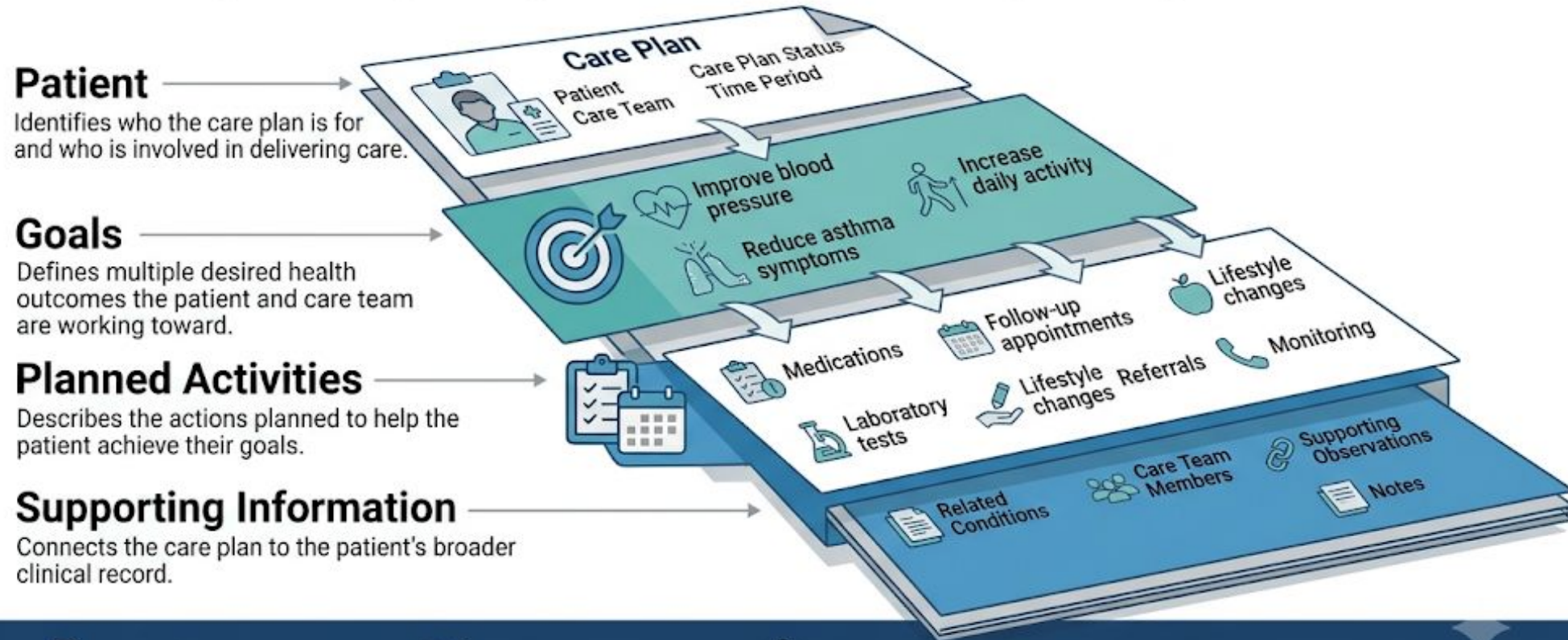
CPG to **Actionable** Care Plans

Writing Care Plans



Inside a FHIR Care Plan

A structured plan that organizes patient-centered care around goals and planned activities.



Patient



Goals



Activities



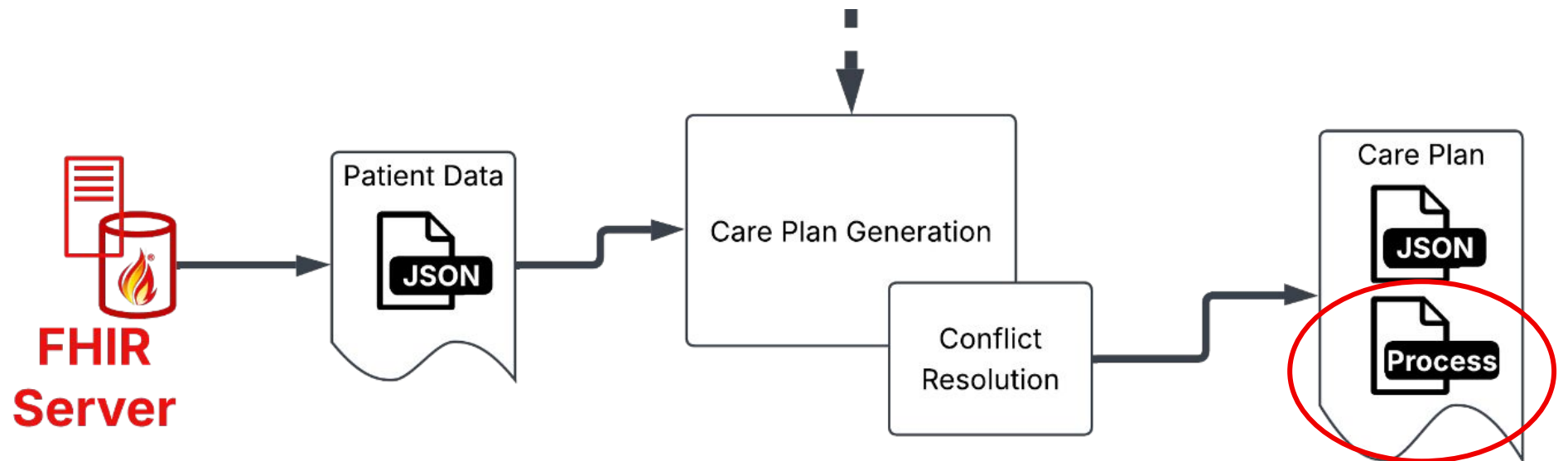
Better Health Outcomes

A FHIR CarePlan provides a standardized, shareable way to coordinate care by linking a patient's goals with the activities needed to achieve them.



CPG to Actionable Care Plans

Process Automation



Business Process Model and Notation (BPMN)

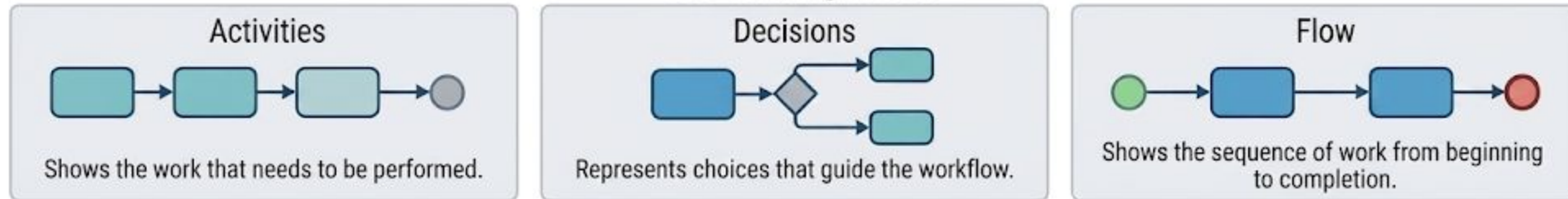
A visual language for describing workflows and business processes.

Why BPMN?



BPMN transforms complex written procedures into clear visual workflows that are easier to understand, review, and improve.

The Building Blocks

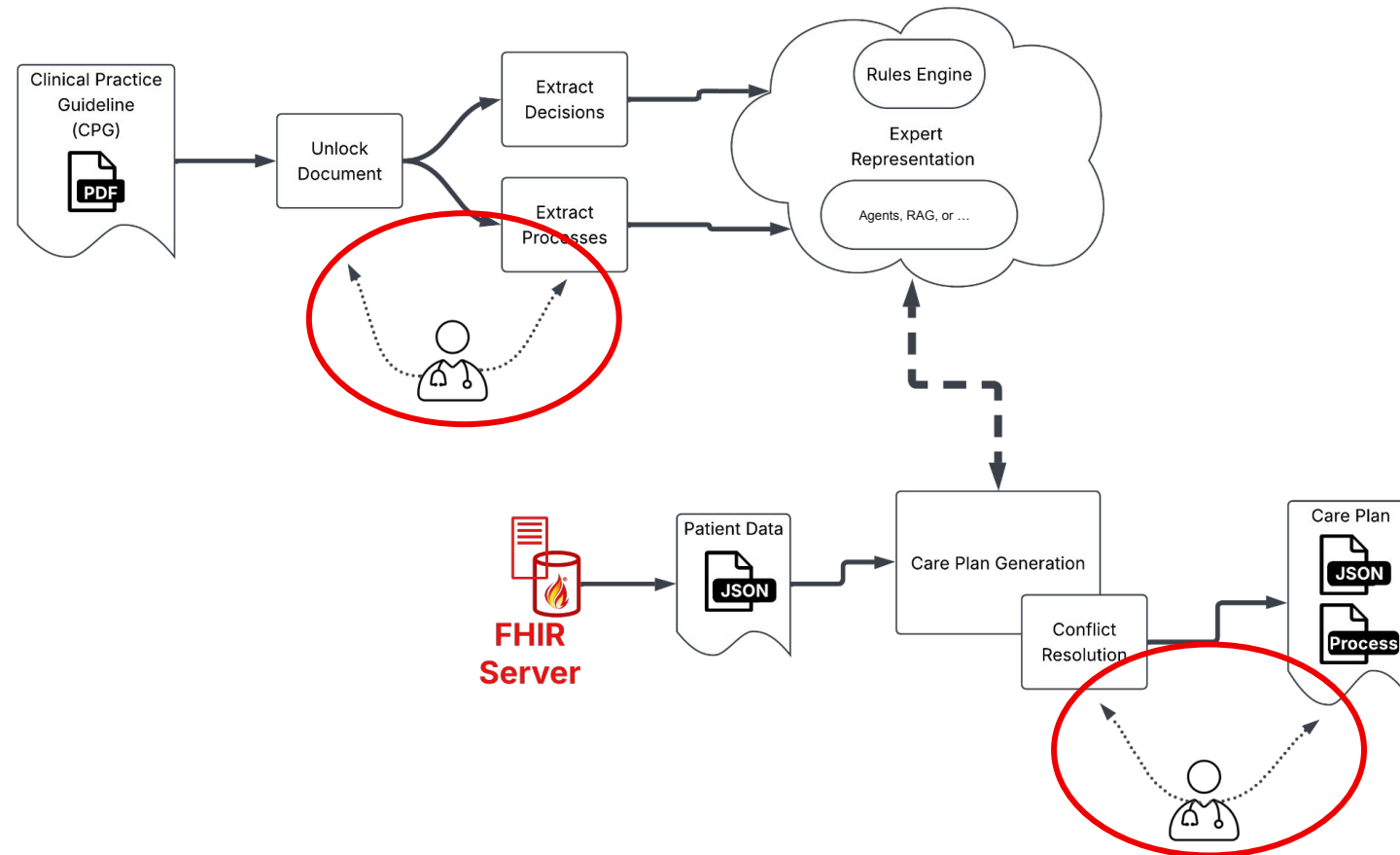


BPMN provides a common visual language that helps subject matter experts, analysts, and developers review workflows together—making complex processes easier to understand, improve, and automate.



CPG to Actionable Care Plans

User Interfaces

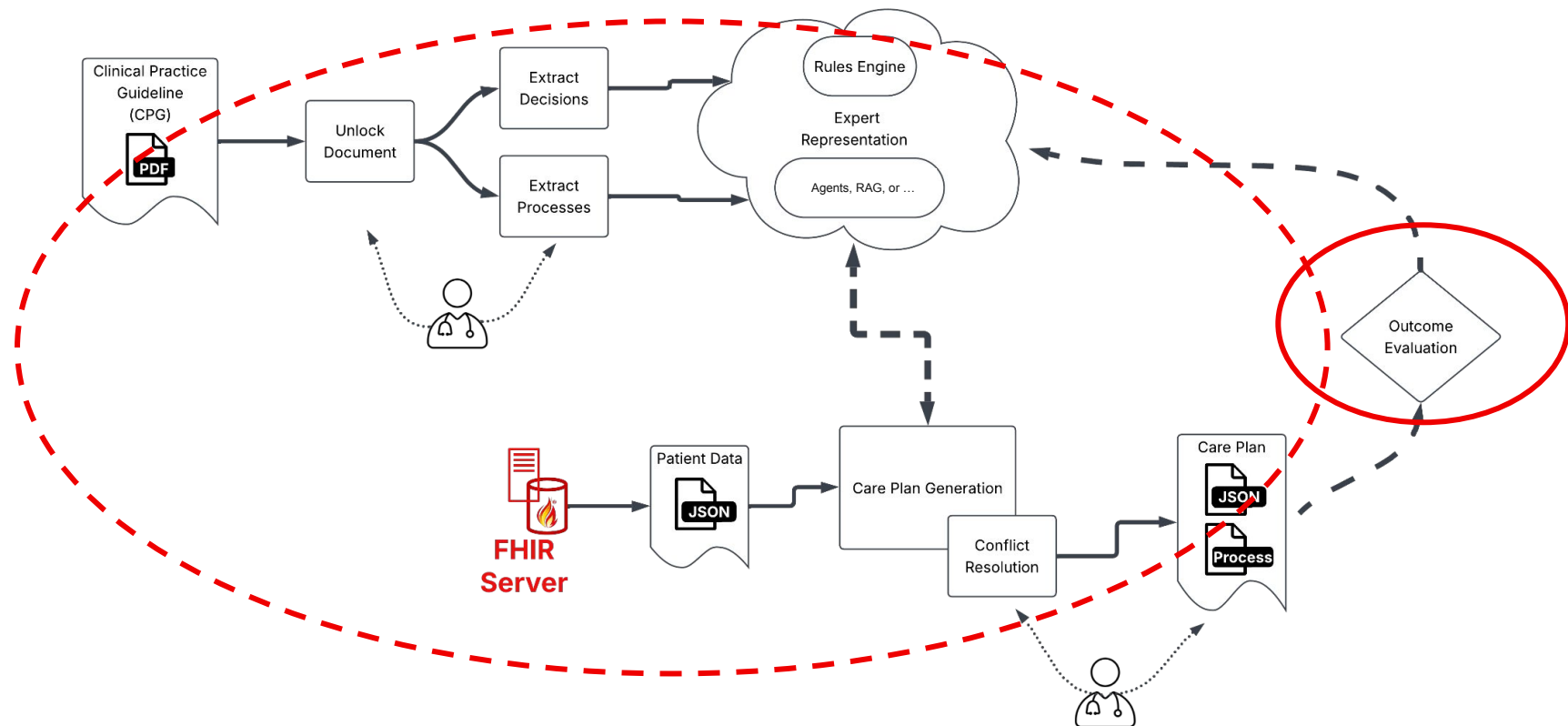


	Upload Input	Review and Edit	Publish
CPG unlocking	PDF (Word and others?)	DMN and Processes from recommendations	To care plan writer: • DMN • Processes
Care Plan writing	IPS (Text and other formats?)	Conflict resolution, goals, activities (BPMN)	To FHIR server and automation engine: • FHIR CarePlan++ Bundle • BPMN



CPG to Actionable Care Plans

Future Work



- ▶ Feedback loops and learning
- ▶ Using smaller models
- ▶ Different ways of representing expert knowledge
- ▶ Importing more data formats, including converting non-FHIR to FHIR to create IPS
- ▶ Scaling all parts of the solution
- ▶ Swappable technology solutions
- ▶ ...





Thank you

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